

Bertha Shipper

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EDUCATION

Vassar College | Poughkeepsie, NY

Expected May 2027

Bachelor's Degree in Computer Science with a Minor in Applied Mathematics

GPA: 3.91/4.00

WORK EXPERIENCE & LEADERSHIP

AI/ML Platform Engineering Intern | *Guardian Life*

May – Aug 2026

- Designed and delivered an **AI observability and governance platform** that unified **AWS CloudWatch** and **Bedrock** telemetry into a trusted analytics system, enabling executive visibility into LLM usage, cost drivers, ownership attribution, and AI governance across **100K+ invocation records** where no centralized reporting previously existed.
- Architected an **end-to-end ingestion pipeline** spanning parsing, normalization, deduplication, enrichment, and persistence, achieving **100% attribution coverage** across users, applications, and agents while improving reporting integrity through identity reconciliation, partitioning controls, and resilient backfill/recovery mechanisms.
- Built a **cache-aware Bedrock cost intelligence engine** and governance framework with pricing refresh logic, lifecycle monitoring, and alerting workflows, providing **100% pricing coverage for 60+ observed models**, delivering actionable cost visibility for executives, and surfacing legacy-model and high-token-consumption risks to support **FinOps**, accountability, and operational decision-making.

Applied Machine Learning Research Intern | *UChicago Data Science Institute* | [GitHub](#)

Jun – Aug 2025

- Built a **scalable multimodal ML pipeline** integrating computer vision, speech, and NLP to predict fMRI brain responses to movie/TV stimuli, enabling efficient analysis of neural data across subjects.
- Developed and tuned **regularized regression models** (ridge/voxel-wise) on an HPC cluster, using cross-validation and automated hyperparameter search; ranked **27/200+** teams in the global 2025 Algonauts Challenge.
- Optimized model performance on high-dimensional data by applying **PCA, statistical modeling**, and **parallelized preprocessing**, cutting compute costs while boosting predictive accuracy.

Product Developer, Treasurer, Co-Founder | *Vassar Ping Pong Club* | [Web Application](#)

Oct 2023 – Present

- Developed and deployed a **full-stack Tournament-Manager web app** to automate event logistics — including dynamic match generation, real-time roster updates, automated leaderboards, and scalable support for 50+ participants — cutting admin work by **80%**.
- Drove **141% growth in club participation** by streamlining registration and improving user experience through automation of tournament logistics and **intuitive web interface design**.
- Manage club finances and budget, allocate funds and distribute resources to maintain financial stability and support member retention.

Lay Review Editor | *Grey Matters Neuroscience Journal* | [Issue 9](#) | [Issue 10](#)

Sep 2024 – Present

- Transform complex neuroscience research into **accessible, reader-friendly articles**, enhancing comprehension while maintaining accuracy and scientific integrity.
- Collaborate with a team of authors, editors, and artists to refine content, streamline revisions, and deliver polished articles on deadline for an acclaimed peer-reviewed publication.
- Manage the editing lifecycle from initial drafts to final publication, consistently meeting deadlines and upholding quality standards.

Data Analysis & Consumer Insights Intern | *Beats* | [Final Presentation](#)

Aug – Sep 2024

- Conducted end-to-end **consumer analytics** in R on large-scale customer feedback datasets, performing data cleaning, feature engineering, and exploratory analysis to uncover behavioral patterns that informed Gen Z-focused marketing strategy across multiple product lines.
- Applied **clustering** and **sentiment analysis** to segment customer populations and quantify satisfaction trends, translating text data into actionable insights that guided campaign targeting and product positioning decisions.
- Designed visualizations with ggplot2 and tidyverse to illustrate demographic, usage, and sentiment trends, enabling marketing and product teams to prioritize features and accelerate campaign planning.

PROJECTS

The Flow State: Neuro-Aesthetic Focus Environment | [Live Web App](#) | [GitHub](#) Feb 2026

- **Motivation:** Traditional productivity timers interrupt focus with notifications and numerical monitoring. I built a closed-loop system that externalizes time perception through generative art and sound, creating a non-intrusive environment designed to support sustained deep work.
- **Impact:** Enabled users to perceive session progress spatially rather than numerically, reducing task-switching and cognitive fatigue while promoting flow-state concentration through perceptual and auditory feedback.
- **Method:** Engineered a real-time generative graphics system that maps elapsed time to progressive visual density; implemented a dual-oscillator Web Audio engine that modulates binaural beats along an arousal curve aligned with cognitive performance research; designed multiple visual logic modes optimized for analytical, creative, and meditative work contexts. Built a peripheral-awareness UI using translucency and low-contrast motion to offload temporal monitoring to peripheral vision and minimize central executive load.
- **Tools:** JavaScript, p5.js, Web Audio API, generative graphics, human-computer interaction, cognitive UX design.

Pantry Pal: Kitchen App for Managing Allergies and Reducing Food Waste | [GitHub](#) Sep 2024 – Jan 2025

★ [People's Choice Award](#), New York Celebration of Women in Computing (NYCWIC) Apr 2025

- **Impact:** Developed an **Android app** that helps users manage pantry inventory, plan meals, and reduce food waste while accommodating dietary restrictions. Improved kitchen organization, prevented duplicate purchases, and enhanced accessibility for users with allergies.
- **Method:** Built on Android platform using **MVC architecture**; implemented dynamic pantry tracking, personalized recipe suggestions, a customizable digital cookbook, and integrated grocery list management. Led product development by defining **user-focused goals**, conducting **usability testing**, and iteratively refining features to optimize **UX, scalability, and reliability**.
- **Tools:** Java, Android Studio, MVC architecture, local storage, user-centered design, interactive UI, usability testing.

Tournament Manager Web Application | [Live Web App](#) | [GitHub](#) Apr 2025

- **Motivation:** My club faced challenges managing large-scale tournaments due to error-prone manual processes, which became unsustainable as participation grew. I aimed to build a **scalable, user-friendly solution** to automate logistics and enhance both management and participant experience.
- **Impact:** Automated tournament logistics, **reducing manual administrative effort by 80%** and enabling smooth management for 50+ participants, contributing to a **141% increase** in club participation.
- **Method:** Developed fair matchup **generation algorithms**, implemented dynamic roster management with real-time score tracking, and iteratively refined features based on **user feedback** and scalability needs. Engineered **persistent data storage using localStorage**, eliminating the need for a backend server. Designed a **clean, responsive UI** optimized for **desktop and mobile**.
- **Tools:** JavaScript, HTML, CSS, responsive web design, full-stack web development.

SKILLS

- **Programming Languages:** Python, Java, R, JavaScript, C, OCaml
- **AI/ML:** Amazon Bedrock, AI Observability, LLM Infrastructure, Machine Learning, NLP, Multimodal Machine Learning, Predictive Modeling, Regression Analysis, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Dimensionality Reduction (PCA), Statistical Modeling
- **Cloud & Platform Engineering:** AWS (Bedrock, CloudWatch), Amazon EKS, Kubernetes, Jenkins, CI/CD, Backend Systems, Middleware Architecture, AI Governance, Data Governance, FinOps, Cost Attribution, Usage Monitoring, Alerting Systems, Linux/Unix, Bash/Shell Scripting
- **Data Engineering & Analytics:** Pipelines, Cleaning & Wrangling, Normalization, Visualization, Sentiment Analysis, Clustering
- **Frameworks & Tools:** PostgreSQL, PyTorch, scikit-learn, NumPy, pandas, Git, GitHub, Jupyter Notebook
- **Software Development:** Full-Stack Web Development, Android Development, OOP, MVC Architecture, UI Design
- **Research & Scientific Computing:** High-Performance Computing (Slurm), Neuroimaging Analysis (fMRI), Cross-Validation, Experimental Design, Statistical Testing